

COMPREHENSIVE TECHNICAL ENCYCLOPEDIA

Non-Invasive Glucose Detection System

Complete Technical Documentation

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1. SYSTEM OVERVIEW

COMPREHENSIVE NON-INVASIVE GLUCOSE DETECTION SYSTEM

End-to-End Architecture: Hardware → Cloud → Clinical Interface

The Non-Invasive Glucose Detection System represents a comprehensive research prototype integrating multi-modal physiological sensors, machine learning algorithms, and clinical decision support interfaces. The system employs a three-layer architecture designed for scalability, reliability, and clinical applicability.

Hardware Platform

- ESP32-S3 DevKitC-1: Dual-core processor with WiFi connectivity
- MAX30102: Dual-wavelength PPG sensor for optical measurements
- TMP117: Medical-grade temperature sensor ($\pm 0.1^{\circ}\text{C}$ accuracy)
- pH Sensor: Laboratory-grade probe for saliva analysis
- ST7789: Color TFT display for real-time visualization

Data Sources & Training

- NHANES 2017-2018: 9,254 participants with demographic data
- D1NAMO Dataset: PPG, ECG, and HRV correlation studies
- Synthetic Features: Physics-based simulation with Monte Carlo sampling

Algorithm Architecture

Model A: Full-Sensor Stacked Ensemble

- Base Learners: Random Forest, Gradient Boosting, SVR
- Meta-learner: Linear regression with cross-validation
- Uncertainty: Quantile regression for 90% prediction intervals
- Performance: $R^2 = 0.8528$, MAE = 12.3 mg/dL, Clarke A+B = 99.2%

Model B: Demographic Risk Classification

- Algorithm: Logistic regression with SMOTE class balancing
- Validation: NHANES-based with 5-fold cross-validation
- Performance: AUROC = 0.73, Precision = 0.68, Recall = 0.71

Performance Analytics

Database & System Integration

- Database: Supabase PostgreSQL with real-time subscriptions
- API: RESTful endpoints with JSON payload transmission
- Frontend: Streamlit dashboard with multi-tab interface
- Security: API key authentication with SSL/TLS encryption

Clinical Validation & Safety

- Out-of-Distribution Detection: Statistical bounds checking
- Clarke Error Grid: 99.2% Zone A+B clinical accuracy
- Uncertainty Quantification: 90% prediction intervals
- Safety Disclaimers: Mandatory research prototype warnings

Technical Specifications

- Sensor Sampling: pH (32 samples), Temperature (10 readings), PPG (500 samples)
- Processing Latency: <2 seconds for feature engineering
- Model Inference: <1 second for predictions
- End-to-End Latency: <5 seconds (button press → result display)

API Reference

Supabase REST Endpoints:

- POST /rest/v1/sensor_readings - Insert sensor data
- GET /rest/v1/sensor_readings?status=eq.pending - Poll pending
- PATCH /rest/v1/sensor_readings?id=eq.{id} - Update predictions

Research Prototype Disclaimer

This system is a research prototype developed for educational and research purposes only. It has not been validated for clinical use and should not be used for medical decision-making. The models are trained on synthetic and limited real-world data and require extensive clinical validation before any potential medical application.